SUNY Old Westbury 2026-2027 Undergraduate Catalog

Course Descriptions

Contract All Courses |

Chemistry, Physics and Physical Science

PX 2000 - Physics 1, Algebra-based

Credits: 3

This is the first course in the non-calculus-based physics sequence, which presents an overview of the fundamental concepts and quantitative methods of physics. This course provides a foundational understanding of classical Newton mechanics, which describes the motion of objects and the forces that cause them to move. Topics include linear and rotational motion, forces and Newton's laws, gravitation, kinetic and potential energy, and momentum.

Prerequisites

MA 2090

or equivalent, completed with grade of "C" or better

When Offered:

Every semester

Note:

Previously named "CP 2220 Structure of Physics I"

PX 2001 - Physics 1, Algebra Based Lab

Credits: 1

This laboratory course provides students with hands-on experiences to complement the theoretical concepts introduced in the corresponding lecture course, PX 2000. Through a series of experiments and simulations emphasizing experimental techniques, data analysis, and error estimation, students will explore fundamental principles of Newtonian mechanics, including motion, forces, energy, and momentum.

Corequisite PX 2000

When Offered:

Every semester

Note:

Previously named "CP 2221 Structure of Physics I Laboratory"

PX 2100 - Physics 2, Algebra-based

Credits: 3

This is the second course in the non-calculus-based physics sequence, which presents an overview of the fundamental concepts and quantitative methods of physics. This course provides a foundational understanding of electricity and magnetism. Topics include electrostatics, electric circuits, magnetostatics, electromagnetic induction, light and electromagnetic waves, and geometric optics.

Prerequisites

PX 2000

or PX 2200

, completed with grade of "C" or better

When Offered:

Every semester

Note:

Previously named "CP 2230 Structure of Physics II"

PX 2101 - Physics 2, Algebra-based Lab

Credits: 1

This laboratory course provides students with hands-on experiences to complement the theoretical concepts introduced in the corresponding lecture course, PX 2100

. Through a series of experiments and simulations emphasizing experimental techniques, data analysis, and error

estimation, students will explore fundamental principles of electricty and magnetism, including electric and magnetic fields, electric circuits and basic electronics, and optics.

Corequisite PX 2100

When Offered:

Every semester

Note:

Previously named “CP 2231 Structure of Physics II Laboratory”

PX 2200 - Physics 1, Calculus-based

Credits: 3

This is the first course in the calculus-based physics sequence, which presents an overview of the fundamental concepts and quantitative methods of physics. This course provides a foundational understanding of classical Newton mechanics, which describes the motion of objects and the forces that cause them to move. Topics include linear and rotational motion, forces and Newton’s laws, gravitation, kinetic and potential energy, and momentum. This course is intended for physics, chemistry, math, and computer science majors, as well as any other science majors and pre-health students who have taken or are enrolled in calculus.

Prerequisites

MA 2090

or equivalent, completed with grade of “C” or better

Corequisite MA 2310

When Offered:

Fall

Note:

Previously named “CP 2240 General Physics I”

PX 2201 - Physics 1, Calculus-based lab

Credits: 1

This laboratory course provides students with hands-on experiences to complement the theoretical concepts introduced in the corresponding lecture course, PX 2200

. Through a series of experiments and simulations emphasizing experimental techniques, data analysis, and error estimation, students will explore fundamental principles of Newtonian mechanics, including motion, forces, energy, and momentum.

Corequisite PX 2200

When Offered:

Fall

Note:

Previously named “CP 2240 General Physics I”

PX 2300 - Physics 2, Calculus-based

Credits: 3

This is the second course in the calculus-based physics sequence, which presents an overview of the fundamental concepts and quantitative methods of physics. This course provides a foundational understanding of electricity and magnetism. Topics include electrostatics, electric circuits, magnetostatics, electromagnetic induction, light and electromagnetic waves, and geometric optics. This course is intended for physics, chemistry, math, and computer science majors, as well as any other science majors and pre-health students who have taken calculus.

Prerequisites

PX 2000

or PX 2200

, completed with grade of "C" or better

When Offered:

Spring

Note:

Previously named "CP 2250 General Physics II"

PX 2301 - Physics 2, Calculus-based Lab

Credits: 1

This laboratory course provides students with hands-on experiences to complement the theoretical concepts introduced in the corresponding lecture course, PX 2300

. Through a series of experiments and simulations emphasizing experimental techniques, data analysis, and error estimation, students will explore fundamental principles of electricty and magnetism, including electric and magnetic fields, electric circuits and basic electronics, and optics.

Corequisite PX 2300

When Offered:

Spring

Note:

Previously named “CP 2251 General Physics II Laboratory”

PX 2700 - Wave Motion: Light and Sound

Credits: 4

An introductory course on the science of waves in light and sound and their applications to modern forms of technology that directly impact our lives, such as wireless communication, image recognition, and remote sensory data collection. Students learn the basic concepts through activities with microcomputer-based laboratory (mbl) technology, designed to explore applications with inquiry-based tasks using sensors and mobile devices to collect and analyze data.

Prerequisites

MA 1020

PX 2900 - Astronomy

Credits: 3

Explore the wonders of the universe in this introductory survey of astronomy. From the night sky to the origin of the universe, this course features evidence-based reasoning, critical thinking, and use of theoretical models and observations. Topics include apparent sky motions, telescopes, planets, structure and evolution of the solar system, stars, black holes, galaxies, and cosmology.

Prerequisites

MA 1020

PX 2901 - Astronomy Laboratory

Credits: 1

This introduction to observational astronomy is the lab associated to PX 2900

. This course addresses the basic techniques of unaided astronomical observing as well as observation with a telescope.

Observations will include the constellations of the night sky, the moon, binary and variable stars, planets, and galaxies. Indoor labs will feature celestial simulations, remote telescopes, and astronomical image processing.

Corequisite PX 2900

PX 3000 - Waves and Modern Physics

Credits: 3

This course provides a study of harmonic oscillation, wave phenomena, special relativity, and an introduction to quantum mechanics. Students will learn to think logically and quantitatively about the behavior of nature and gain a systematic approach to problem solving, preparing them for advanced study of classical and quantum mechanics.

Prerequisites

PX 2200

or PX 2000

with grade of "C" or better, MA 2310

When Offered:

Fall of odd-numbered years

PX 3100 - Mathematical Methods in Physics

Credits: 3

This course covers a variety of essential mathematical concepts and techniques used in physics, focusing on applications to physical problems rather than on deviations and proofs. Topics will include infinite series, complex functions, differential equations, vector spaces and linear algebra, and Fourier analysis.

Prerequisites

MA 2320

When Offered:

Fall of odd-numbered years

Note:

This course is cross-listed with CP 3230 Mathematical Methods in the Physical Sciences

PX 3200 - Computational Physics

Credits: 3

This course introduces techniques and provides the foundational knowledge and computing skills needed to work in a modern physics laboratory environment. Students are introduced to commonly used computing systems and networking, the basics of programming using Python, numerical methods for modeling complex systems, and analog circuits. The topics covered include simple harmonic motion, waves and sound, and RLC circuits.

Prerequisites

PX 2200

or PX 2000

with grade of "C" or better, MA 2310

When Offered:

Fall of even-numbered years

PX 3600 - Mechanics

Credits: 3

This course presents the basic ideas and techniques of analytical mechanics, in particular the calculus of variations, Hamilton's Principle, and Lagrangian dynamics. Topics include drag, oscillations, gravity and central force motion, symmetries and conservation laws, and chaos.

Prerequisites

PX 3000

with grade of "C" or better, MA 2320

When Offered:

Spring of even-numbered years

PX 3700 - Electromagnetism

Credits: 3

This course covers the principles and techniques of electromagnetic theory. Topics include electrostatics and magnetostatics, electric and magnetic fields in matter, electromagnetic induction, Maxwell's equations, and electromagnetic radiation.

Prerequisites

PX 2300

or PX 2100

with grade of "C" or better, MA 3330

When Offered:

Spring of odd-numbered years

PX 3800 - Relativity

Credits: 3

This course presents the concepts and principles of Einstein's theory of special relativity, as well as an introduction to general relativity. Topics include relativistic kinematics and mechanics, Minkowski spacetime, Lorentz transformations, four-vectors, the metric, and spacetime curvature. Applications to particle physics, astrophysics, and cosmology are covered.

Prerequisites

PX 3000

with grade of "C" or better, MA 2320

When Offered:

As needed

PX 3900 - Experimental Physics and Electronics Lab

Credits: 3

This course provides a hands-on introduction to the electronics and instrumentation used in experimental physics. Topics include DC and AC circuits, filters, power supplies, transistors, diodes, operational amplifiers, and analog to digital converters and logic. This is primarily a laboratory class, but time will also be spent in and out of the class on the mathematical description of the circuits and instruments, as well as discussions of the theory underlying modern components, such as the diode and transistor.

Prerequisites

PX 2301

or PX 2101

with grade of "C" or better, MA 2320

When Offered:

Spring of odd-numbered years

PX 4100 - Thermodynamics

Credits: 3

An introduction to the concepts and laws of thermodynamics and their statistical basis. Topics will include temperature, heat, thermal equilibrium, reversible and irreversible processes, entropy, and free energy. Applications such as ideal gases, heat engines, refrigerators, and phase transitions will be covered.

Prerequisites

PX 2300

or PX 2100

completed with a grade of "C"or better; CP 2130

completed with grade of "C" or better; MA 2320

completed with a grade of "C" or better,

When Offered:

Every Fall.

Note:

This course is cross-listed with CP 4700 Physical Chemistry 1

PX 4200 - Quantum Mechanics

Credits: 3

This course is an introduction to the concepts and methods of quantum mechanics. Topics will include the postulates of quantum mechanics, Schrödinger's equation, Dirac notation, measurements, operators and observables, angular momentum and spin, and the exclusion principle. Applications will include harmonic oscillator, the hydrogen atom, and scattering.

Prerequisites

PX 3000

with grade of "C" or better, MA 2320

When Offered:

Spring of even-numbered years

PX 4300 - Applied Quantum Mechanics

Credits: 3

This course is a survey of the basic principles of quantum mechanics with a focus on applications in modern, nuclear, and molecular physics. Topics include an overview of quantum mechanics, the hydrogen atom, electronic structure, and the quantum theory of atoms and molecules. Emphasis is placed on electronic structure, spectroscopy, and computational methods. Theoretical concepts are applied to real-world problems in physics, chemistry, and materials science throughout the semester.

Prerequisites

CP 2130

completed with a grade of "C" or better; MA 2320

completed with a grade of "C" or better, PX 2300

or PX 2100

completed with a grade of "C" or better.

When Offered:

Spring of even-numbered years

Note:

This course is cross-listed with CP 4710 Physical Chemistry II.

PX 4600 - Applied Mechanics

Credits: 3

This course explores statics and dynamics of rigid bodies and of stress and strain applied to "real" materials with elasticity and finite strength. Topics include simple equilibrium, structural analysis, inertia tensors, centroids, energy and momentum, and strength of materials, as well as, concepts of stress and strain in structural members experiencing tension, compression, shear, bending, and twisting.

Prerequisites

PX 3000

with grade of "C" or better, MA 2320

PX 5000 - Advanced Topics in Physics

Credits: 3

This course covers selected advanced topics. Popssible areas of study may include quantum mechanics, particle physics, condensed matter, biophysics, astrophysics, cosmology, or string theory.

Prerequisites

Permision of the instructor

When Offered:

As needed

PX 5100 - Econophysics

Credits: 3

Econophysics is the application of ideas from statistical physics to the financial markets. In this course, the dynamics of markets are analyzed from a physics perspective. The statistical methods needed to model financial markets are introduced: classes of Brownian motion, probability distributions and limit theorems. Comparison with other physical systems are used to develop intuitions for these methods. Stochastic models for price dynamics and time-dependent models for equity valuations, such as the Black-Scholes equation, used in options pricing, are developed. Real world markets, such as equity stock markets, fixed income markets, and commodities markets, are considered.

Prerequisites

PX 4100

, BU 4762

, CS 2510

When Offered:

As needed

PX 5500 - Senior Seminar 1

Credits: 1

The first semester of a year-long capstone course for physics majors. Students engage in independent research projects under the guidence of faculty mentors, culminating in a written article, a poster presented at Research Day, and an oral presentation to faculty and peers. The course covers literature reviews, scientific writing, and effective presentation skills.

Prerequisites

Senior standing or permission of the instructor

When Offered:

Fall Only

PX 5600 - Senior Seminar 2

Credits: 1

The second semester of a year-long capstone course for physics majors. Students engage in independent research projects under the guidence of faculty mentors, culminating in a written article, a poster presented at Research Day, and an oral presentation to faculty and peers. The course covers literature reviews, scientific writing, and effective presentation skills.

Prerequisites

PX 5500

When Offered:

Spring Only

PX 5900 - Research in Physics

Credits: 2-4

A research project carried out under the supervision of a faculty member. Each project includes a survey of the appropriate literature, a theoretical or experimental investigation, and a comprehensive report of the results.

Prerequisites

Permission of the physics faculty

When Offered:

Every Semester

PX 9990 - Independent Study in Physics

Credits: 2-4

An academic project, other than research, carried out under the supervision of a faculty member.

Prerequisites

Permission of the physics faculty

When Offered:

Every Semester

Contract All Courses |